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1. What type of TV screen technologies are currently available in Australia and which are the most frequent?

1.1 What data is needed?

Column Name	**Data type**	**Description**
Screen_Tech	String	Type of TV technology (LCD, LED, OLED)
Availability Status	Categorical	Filter to include only Available models.
Count(Model_No)	Numeric (integer)	Frequency/count of available models for each Screen_Tech (used to find the most frequent).

The relevant fields are **Screen_Tech** and **availability** (records where **Availability Status = Available**). The frequency (model count) for each **Screen_Tech** is then used to determine the most frequent technology.

1.2 What chart type would be good to represent this data and give some reasons?

A **bar chart** is suitable because it compares **discrete categories** by their **counts**, making the highest bar the most frequent technology

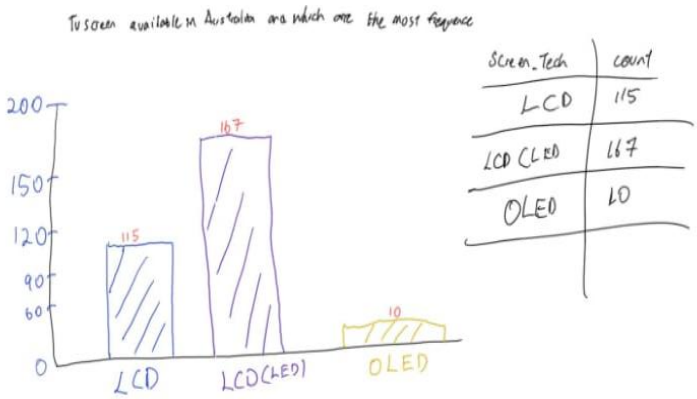


Figure 1.2.1 : Sketch A bar chart about **Screen_Tech** available in Australia and which are the most frequent?

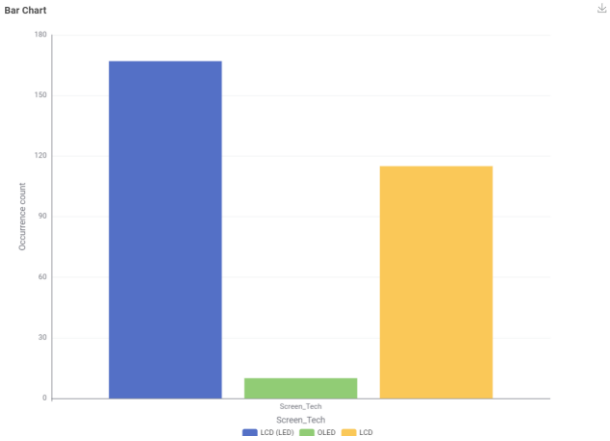


Figure 1.2.2: Knime chart about **Screen_Tech** available in Australia and which are the most frequent?

1.3 KNIME workflow, data table and chart

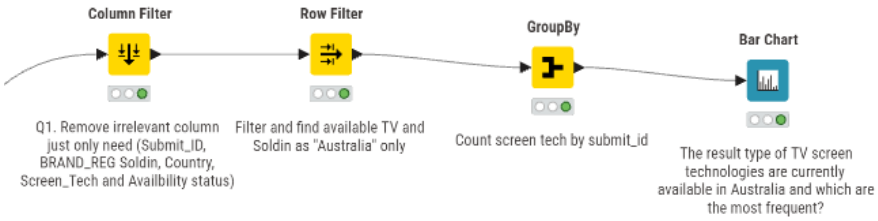


Figure 1.3.1 KNIME Workflow for Screen Technology Analysis

#	RowID	Screen_Tech String	Availability Status String	Count*(Submit_ID) Number (Integer)
1	Row0	LCD	Available	115
2	Row1	LCD (LED)	Available	167
3	Row2	OLED	Available	10

Figure 1.3.2 table displays the number of available models for each type of screen technology.

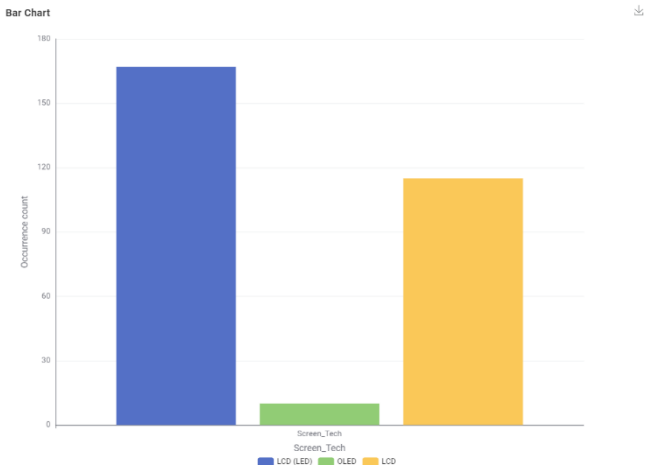


Figure 1.3.3 Bar Chart of Screen Technology Distribution

2. What screen sizes are currently available, and which are the most frequent?

2.1 What data is needed?

Column Name	**Data type**	**Description**
screensize	Numeric	Size of the TV screen (e.g., 55, 65 inches)
Availability Status	Categorical	Filter to include only Available models.
Frequency (count of Available)	Numeric (integer)	Number of available models for each screensize (used to find the most frequent).

the relevant fields are **screensize** and **Availability Status**. Counting the number of **Available** models for each **screensize** produces the frequency distribution needed to identify the most frequent sizes.

2.2 What type of chart and data table?

A **bar chart** is appropriate because screen sizes can be treated as categories and the most frequent size(s) are shown by the tallest bars.

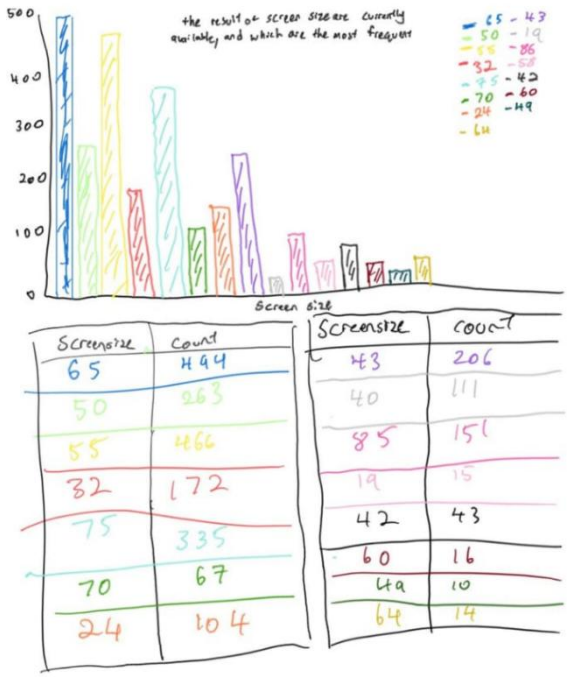
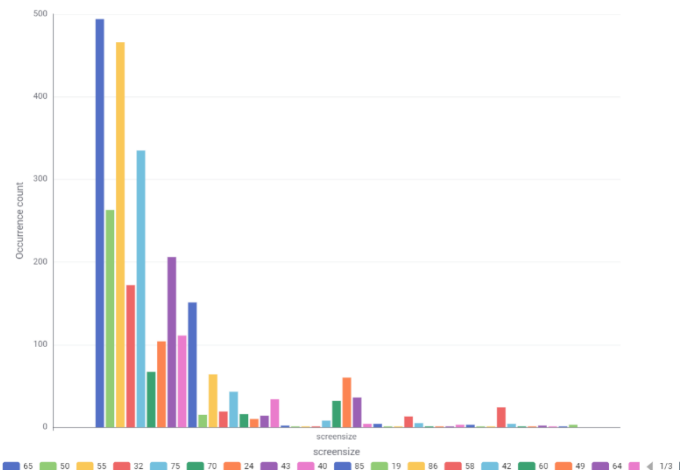
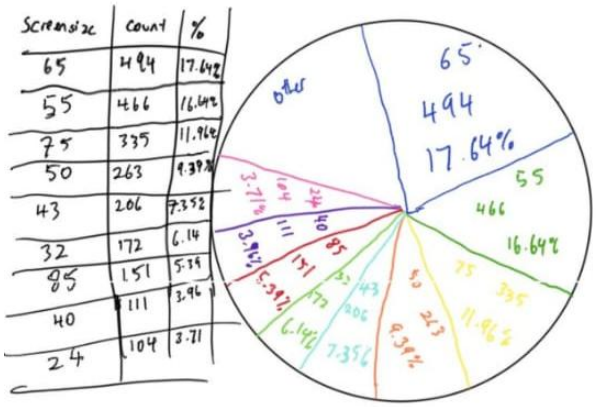


Figure 2.2.1 Sketch A bar chart about screen sizes are currently available, and which are the most frequent?



Pie (small number of categories, will need to convert to percent)



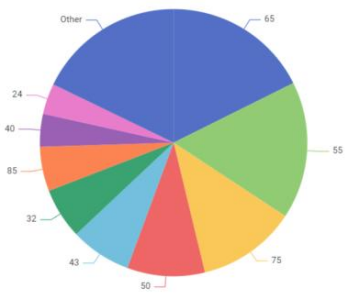


Figure 2.2.2 Sketch A Pie chart about screen sizes are currently available, and which are the most frequent?

2.3 KNIME workflow, data table and chart

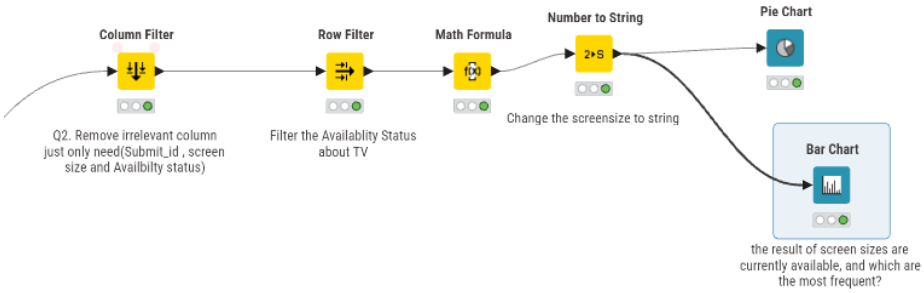


Figure 2.3.1: KNIME Workflow for Screen Size Analysis

R...	screensize String	Availability Status String	Count*(Submit_ID) Number (Integer)
Row0	100	Available	24
Row1	115	Available	4
Row2	116	Available	1
Row3	16	Available	3
Row4	18	Available	1
Row5	19	Available	15
Row6	21	Available	1
Row7	22	Available	5

Figure 2.3.2: Data Table of Screen Size Frequency

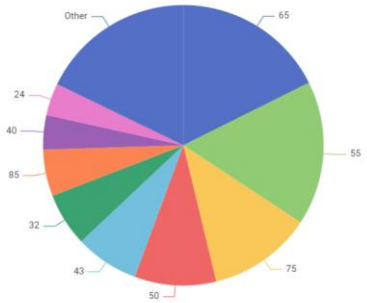


Figure 2.3.3: Bar Chart of Screen Size Distribution

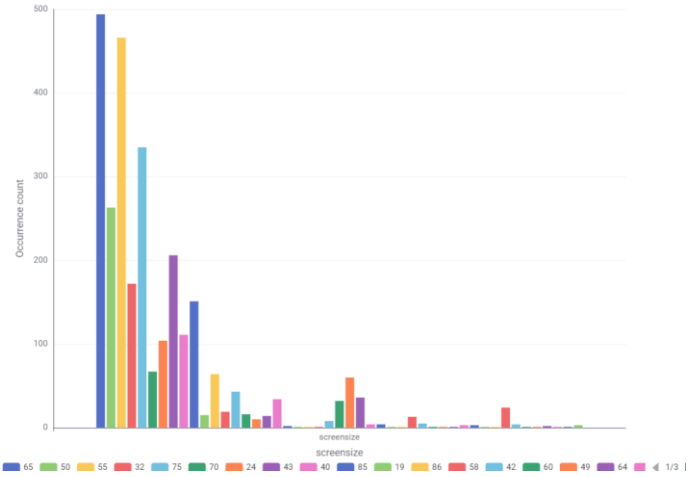


Figure 2.3.4: Pie Chart of Screen Size Distribution

3. Which brands have the greatest number of different models?

3.1 What data is needed?

Column Name	**Data type**	**Description**
Brand_Reg	String	Name of the TV manufacturer
Count(Model_No)	Numeric (integer)	Number of distinct models for that brand (used to rank brands).

the required fields are **Brand_Reg** and **Count(Model_No)** (already aggregated as model variety per brand). Ranking brands by **Count(Model_No)** identifies which brand has the most different models.

3.2 What type of chart and data table?

A **bar chart** works best because it supports clear **ranking** and improves readability when brand names are long.

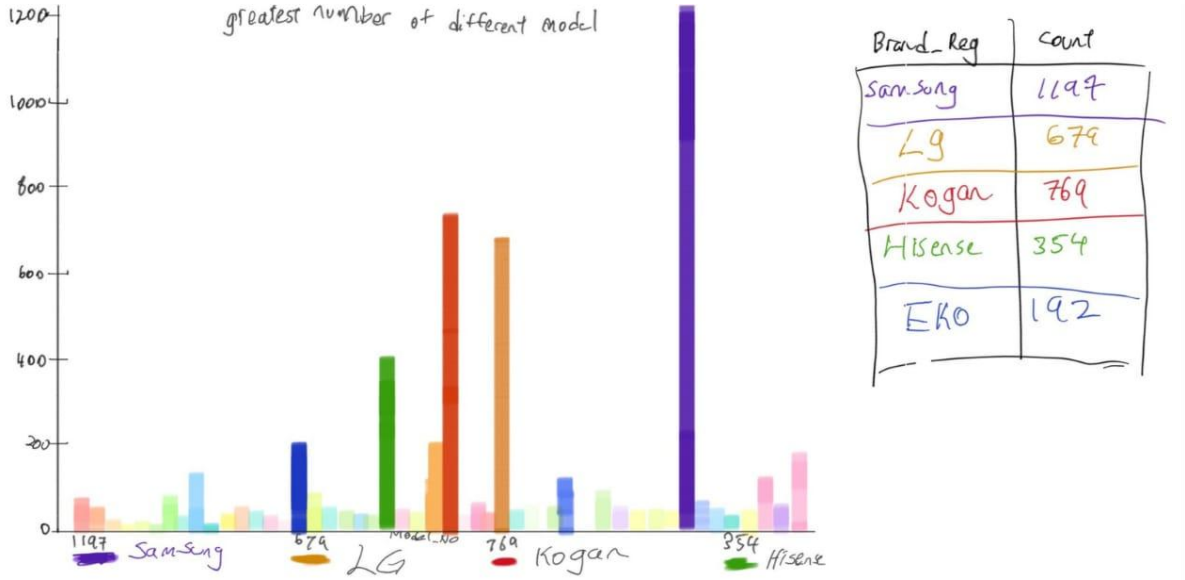


Figure 3.2.1 Sketch A bar chart about which brands have the greatest number of different models?

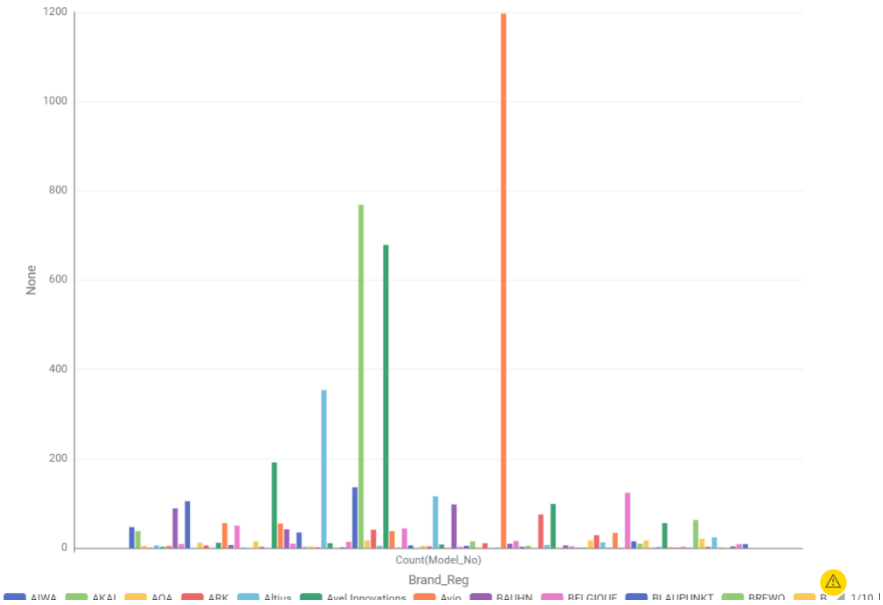


Figure 2.3.4: Pie Chart of Screen Size Distribution

3.3 KNIME workflow, data table and chart

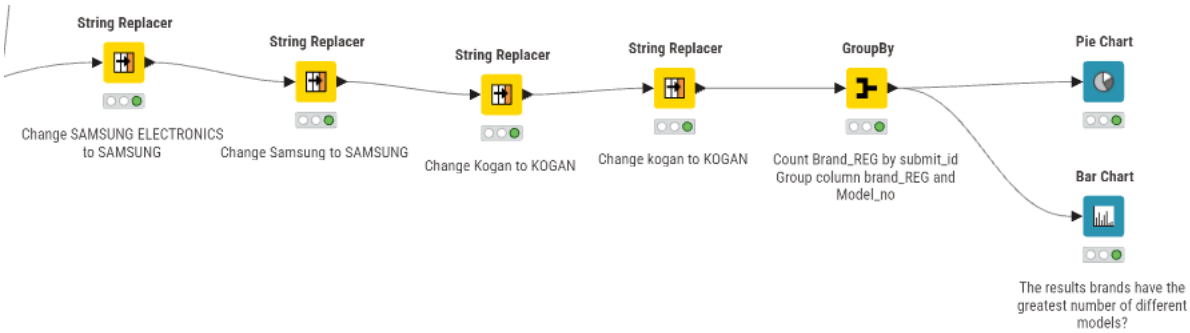


Figure 3.3.1: KNIME Workflow for Brand Analysis

RowID	Brand_Reg T String	Count(Model_No) 123 Number (Integer)
Row0	AIWA	47
Row1	AKAI	38
Row2	AOA	5
Row3	ARK	1
Row4	Altius	6
Row5	Avel Innovations	3
Row6	Avio	5
Row7	BAUHN	89
Row8	BELGIQUE	9
Row9	BLAUPUNKT	105

Figure 3.3.2 Data Table of Brand Model Counts

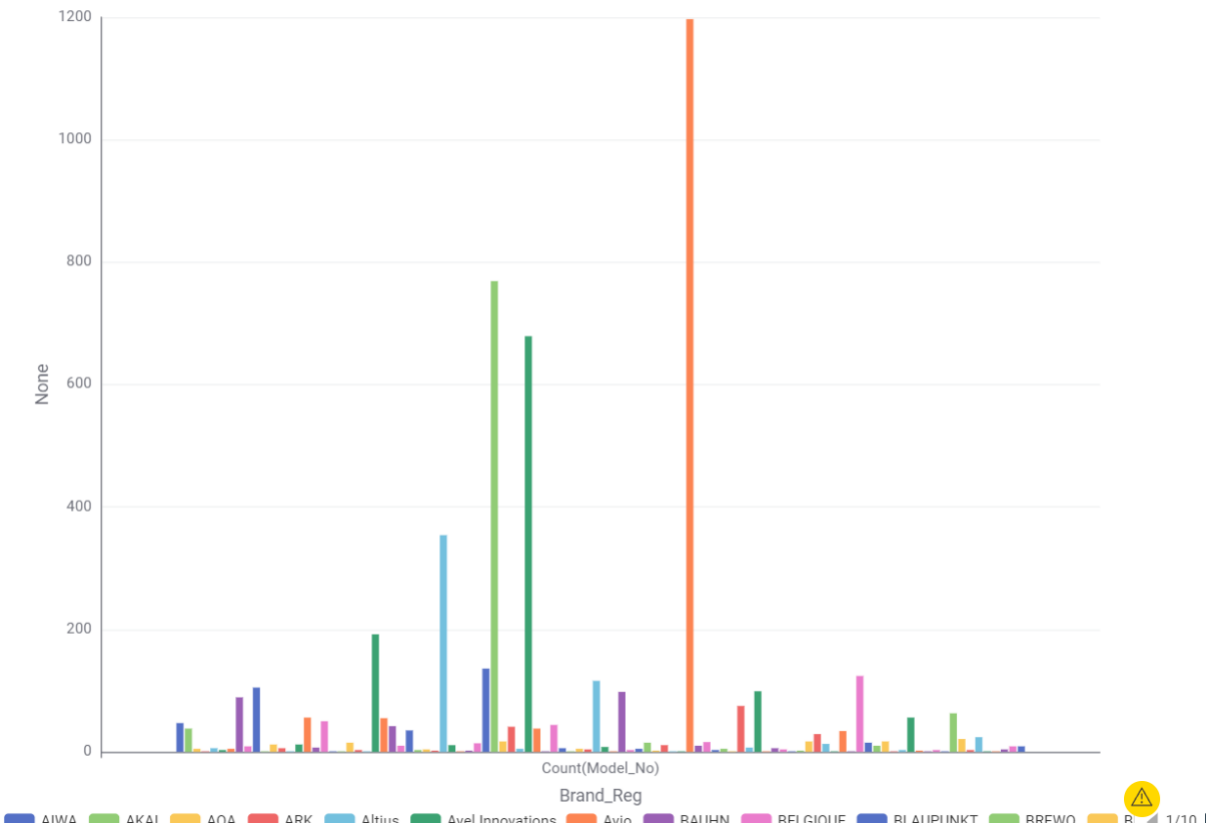


Figure 3.3.3 Bar Chart of Brand Comparison

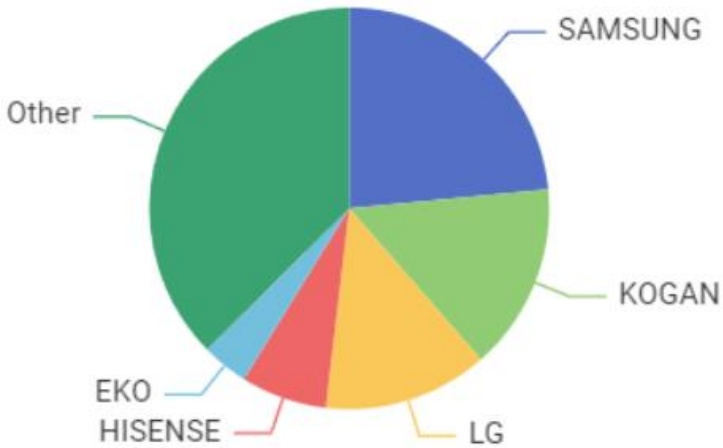


Figure 3.3.4 Pie Chart of Brand Comparison

4. Which type of screen technology consumes the least amount of power?

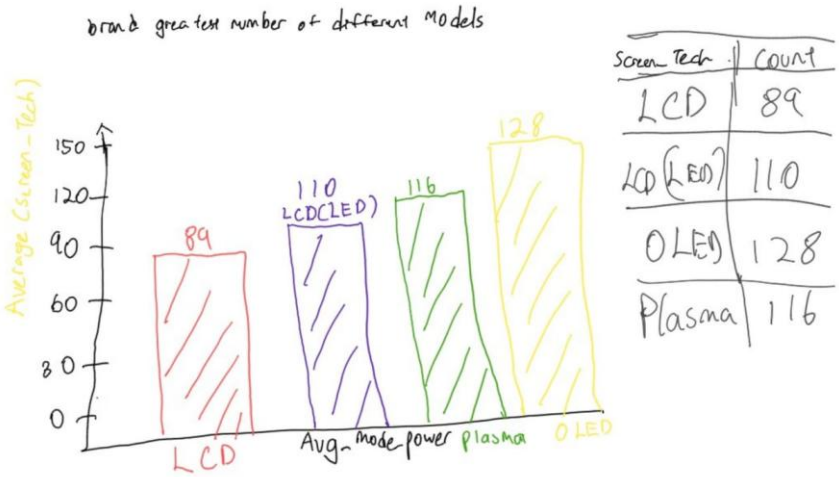
4.1 What data is needed?

Column Name	**Data type**	**Description**
Screen_Tech	String	Type of display technology
Avg_mode_power	Numeric	Power consumption of the TV (Watts)
Count*(Submit_ID)	Numeric (integer)	Number of available submissions contributing to the mean (useful for context/weighting).

The fields **Screen_Tech** and **Mean(Avg_mode_power)** provide the average power consumption per technology category. The screen technology with the smallest **Mean(Avg_mode_power)** is the least power-consuming.

4.2 What type of chart and data table?

A **bar chart** is appropriate because it compares **aggregated numeric power values** across discrete technologies, with the smallest bar indicating the lowest power use



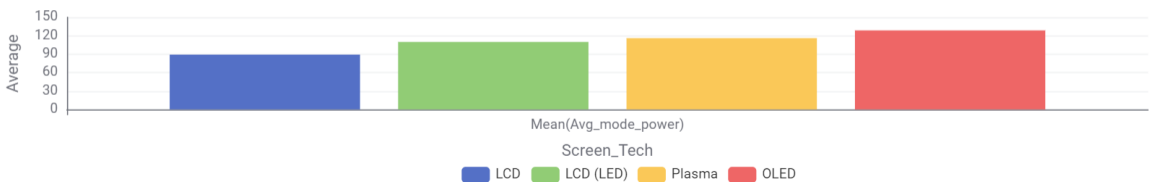


Figure 4.2.1 Sketch A bar chart about screen technology consumes the least amount of power

4.3 KNIME workflow, data table and chart

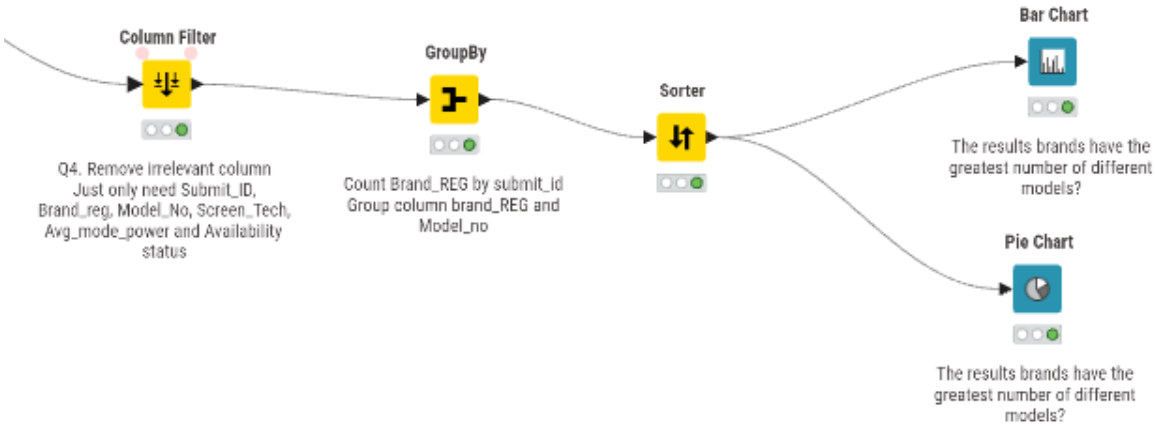


Figure 4.3.1 KNIME Workflow for Power Consumption Analysis

RowID	Screen_Tech String	Count*(Submit_ID) Number (Integer)	Mean(Avg_mode_power) Number (Float)
Row0	LCD	420	89.283
Row1	LCD (LED)	2332	109.928
Row3	Plasma	6	116.167
Row2	OLED	283	128.757

Figure 4.3.2 Data Table of Average Power Consumption

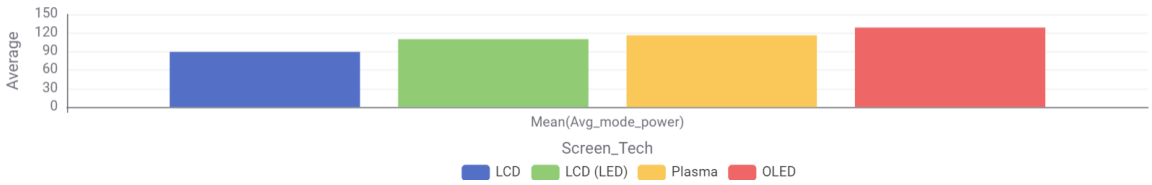


Figure 4.3.3 Bar Chart of Power Consumption by Technology

5. What is the relationship between screen size and power use?

5.1 What data is needed?

Column Name	**Data type**	**Description**
screensize	Numeric	Size of TV screen

Labelled energy consumption (kWh/year)	Numeric	Power consumption (Watts)
Count*(Submit_ID)	Numeric (integer)	How many records correspond to the point (can be used to size/opacity points).

the paired variables are **screensize** (x-axis) and **Labelled energy consumption (kWh/year)** (y-axis). Using these paired values to plot all points allows the overall pattern between size and energy consumption to be observed.

5.2 What type of chart and data table?

A **scatter plot** is the best choice because it visualises the **relationship between two quantitative variables** and reveals trends, clustering, and spread.

Scotter plot (easiest to see differences)

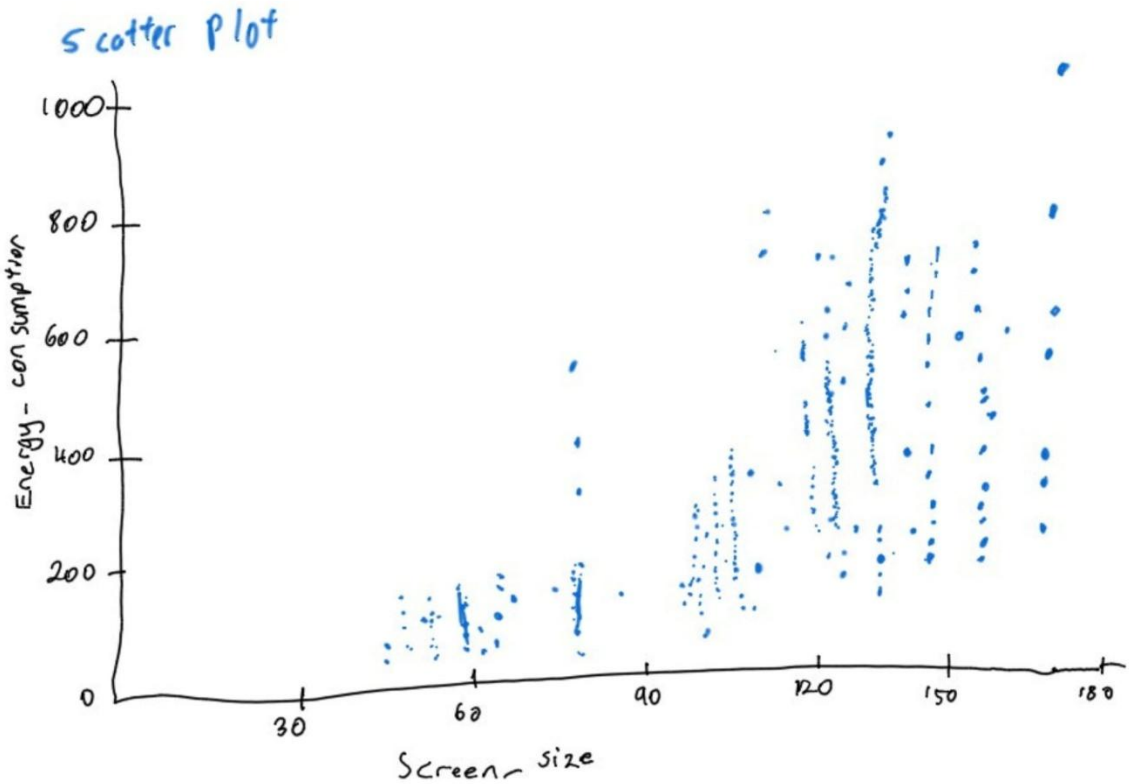


Figure 5.2.1 Sketch A Scotter plot about the relationship between screen size and power use

5.3 KNIME workflow, data table and chart

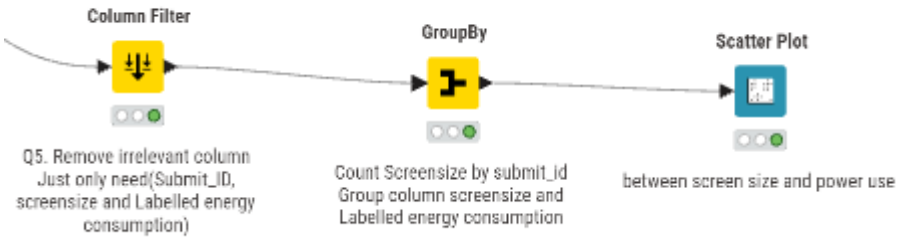


Figure 5.3.1 KNIME Workflow for Size vs Power Analysis

#	RowID	screensize Number (Float)	Labelled energy consumption (kWh/year) Number (Float)	Count*(Submit_ID) Number (Integer)
1	Row0	39.486	58	1
2	Row1	39.487	56	1
3	Row2	39.487	58	2
4	Row3	39.491	51	1

Figure 5.3.2 Data Table of Screen Size vs Power

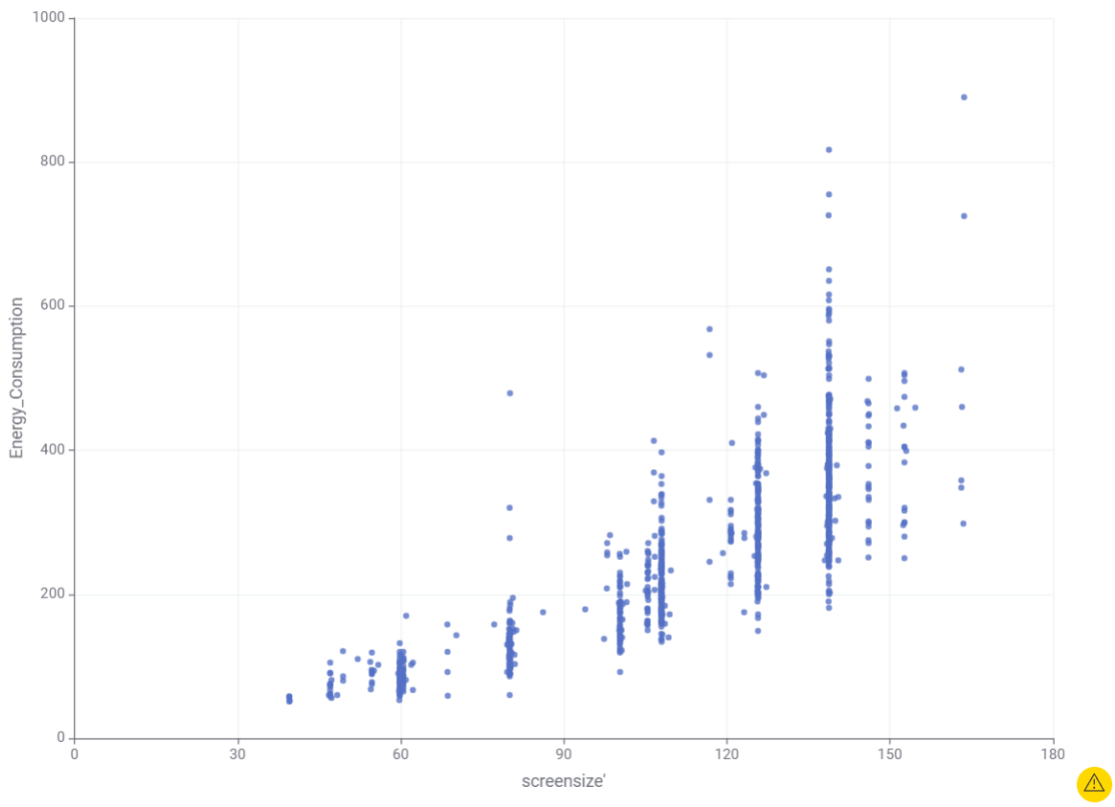


Figure 5.3.3 Scatter Plot of Screen Size vs Power Use

6. What is the relationship between star rating and screen size?

6.1 What data is needed?

Column Name	**Data type**	**Description**
screensize	Numeric	Size of TV screen
Star	Numeric (decimal)	Energy efficiency rating of the TV
Count (Submit_ID)	Numeric (integer)	Number of records supporting the (size, star) combination (can be used to size/opacity points).

the paired variables are **screensize** (x-axis) and **Star** (y-axis). Plotting these values allows assessment of whether star ratings change across screen sizes.

6.2 What type of chart and data table?

A **scatter plot** is suitable because it shows the **association (or lack of association)** between two variables and makes any trend or concentration of high/low ratings easy to see.

Scotter plot (easiest to see differences)

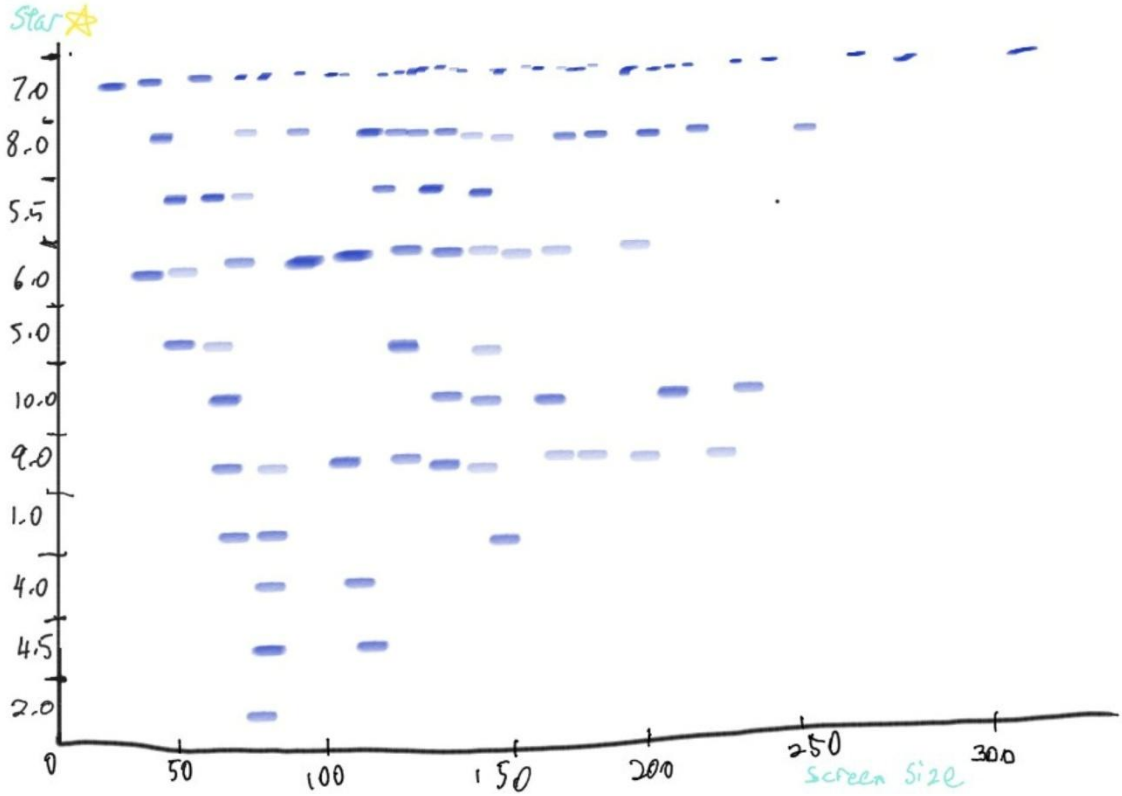


Figure 6.2.1 Sketch A Scotter plot about the relationship between star rating and screen size

6.3 KNIME workflow, data table and chart

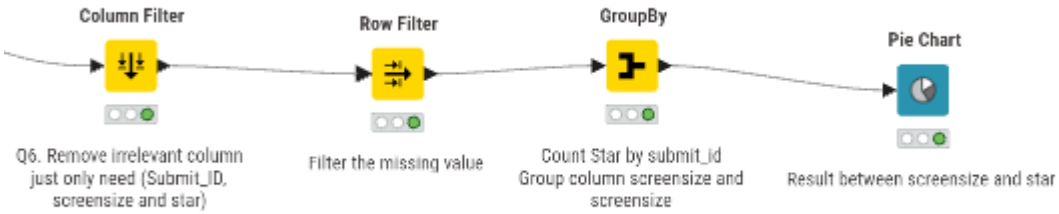


Figure 6.3.1 KNIME Workflow for Star Rating Analysis

RowID	screensize Number (Float)	Star String	Count*(Submit_ID) Number (Integer)
Row0	39.486	7.0	1
Row1	39.491	8.0	1
Row2	47.013	5.5	1
Row3	47.013	6.0	3

Figure 6.3.2 Data Table of Screen Size vs Star Rating

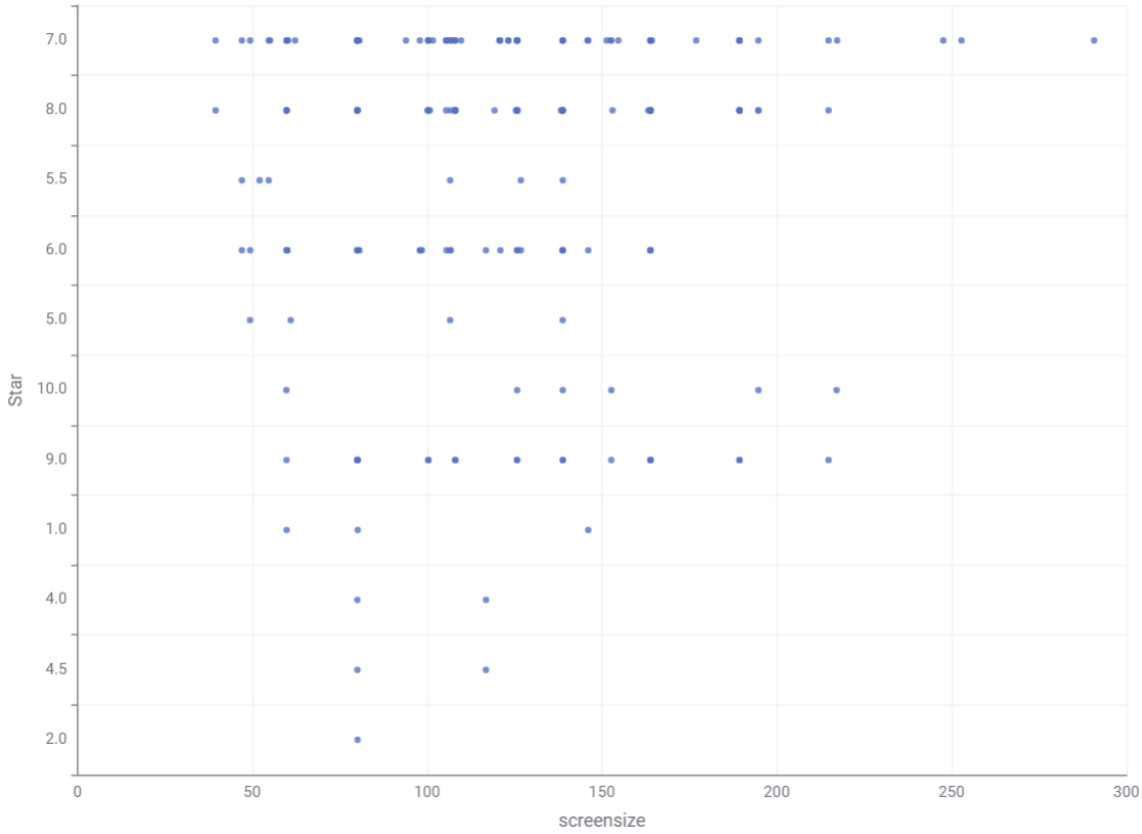


Figure 6.3.3 Scatter Plot of Star Rating vs Screen Size